

Free Fall – Data Sheet

Name _____ Group _____

Table A

Trial	Acceleration (m/s ²)		
1			
2			
3			
4			
5			
6		Average: $\bar{x}$	
7		% Error of Average	
8		Standard Deviation: sd	

Δx is the spacing between the leading edges. $\Delta x =$ _____

Table B

Raw Data (from computer)	Time at each Leading Edge	Midpoint of Time Interval	Average Velocity for Time Interval	Position of Each Leading Edge
Δt (s)	t (s)	t_{mid} (s)	v_{ave} (m/s)	x (m)
	0			0

Δt	t_i	$t_{\text{mid}} = \frac{t_i + t_f}{2}$	$v_{\text{ave}} = \frac{\Delta x}{\Delta t}$	x_i
	$t_f = t_i + \Delta t$			$x_f = x_i + \Delta x$

Question 1: Include a graph made using software of x vs. t (from *your* calculations in the Table B. This is from the x column and the t column.) The graph should be a parabola. From the equation of the best-fit parabola find the acceleration and the initial velocity.

Question 2: Also include a graph v vs. t (from *your* calculations of v_{ave} and t_{mid} .) draw the best-fit line and find its slope. Compare the slope with the accepted value of 9.80 m/s^2 ; give the % error.

Question 3: From your graph of v vs. t find the speed of the picket fence when the first leading edge passed the photogate.

Question 4: If the picket fence were dropped at an angle how would this affect your results? (The picket fence is in a vertical plane but its side is skewed from vertical, as shown.) Would your measured value of g be too large or too small? What percent error would be caused by dropping the picket fence at a 5° angle from vertical? At a 10° angle?

